Shizhao Gao

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ORCID: 0009-0009-1131-3735 | Expected M.Eng. completion: June 2027

Research Profile

M.Eng. candidate researching efficient video-language understanding, test-time scaling, long-video reasoning, and temporal representation learning. My work combines inference-cost analysis, reproducible evaluation, and self-supervised video modeling. My research aims to improve the accuracy and computational efficiency of video-language models through adaptive test-time computation.

Education

Fujian Agriculture and Forestry University Sep. 2024–Jun. 2027 (expected)

M.Eng. in Artificial Intelligence | Top 10% in comprehensive assessment.

Joint research training at the Quanzhou Institute of Equipment Manufacturing, Haixi Institute, Chinese Academy of Sciences. Supervisors: Prof. Jun Li and Assoc. Prof. Qiming Li.

Thesis: Research on Weakly Supervised Repetitive Action Counting Methods Based on Action Similarity.

South-Central Minzu University Sep. 2018–Jun. 2022

B.Eng. in Software Engineering | Top 10% in comprehensive assessment.

Publications – Published / Accepted

[1] **Shizhao Gao**, Jun Li, Qiming Li. "Count What Repeats: Period-Adaptive Multi-Scale Consistency for Self-Supervised Repetitive Action Counting." *CVPR 2026 Findings*.

[2] Jun Li, **Shizhao Gao**, Tianwen Hu, Yinhui Xie, Zhouhua Liu, Qiming Li. "TWCRAC: Leveraging Temporal Cycle Consistency and Dynamic Thresholding for Robust Repetitive Action Counting." *Pattern Recognition*, accepted.

Selected Research Experience

Test-Time Scaling for Efficient Video-Language Understanding 2026

First Author | *Manuscript in Preparation*

- Investigate the accuracy–cost trade-off in video question answering using Qwen3-VL-8B, with budget analysis, ablation studies, and transfer evaluation on Qwen2.5-VL-7B.
- Evaluated on **9,015 questions across five video datasets**. Improved accuracy by **2.11 percentage points** over a query-adaptive keyframe baseline while using **72.23% fewer input tokens**.
- Improved accuracy by **3.15 percentage points** over uniform sampling at comparable input-token cost, demonstrating stronger video understanding within a similar inference budget.

Frame Sufficiency and Adaptive Routing for VideoLLMs 2026

Independent research | *Open-source code and evaluation artifacts*

- Studied minimum sufficient frame budgets on **1,336 LongVideoBench validation examples** with Qwen2.5-VL and LLaVA-Video; implemented visual-token masking, answer-option scoring, and confidence-based routing.
- Saved **18.0% FLOPs** for Qwen at near-baseline accuracy under an independent-forward-pass cost model. Found that the gain did not transfer to LLaVA, establishing a cross-model limitation.
- Released model adapters, automated tests, CI, experiment artifacts, and data/licensing documentation for reproducible evaluation.

Self-Supervised Counting, Action Quality Assessment and Anomaly Detection 2024–Present

First author / lead contributor

- Led problem formulation, period-adaptive temporal modeling, PyTorch implementation, and ablation studies. Achieved **0.129 MAE / 0.842 OBO** on RepCount and **0.098 MAE / 0.879 OBO** on Countix; published in CVPR 2026 Findings.
- Extended the research to action quality assessment and abnormal-action detection; established an industry collaboration to advance technical validation and application in real business settings.

Manuscripts Under Review

[3] **Shizhao Gao**, Jun Li, Qiming Li. "Semantic-Guided Orthogonal Decoupling and Adaptive Signal Tuning for Self-Supervised Repetitive Action Counting." Under review at *International Journal of Computer Vision*.

[4] **Shizhao Gao**, Jun Li, Yinhui Xie, Qiming Li. "Nano-TCC: A Resource-Aware On-Device Repetition Sensing System for Mobile Edge Devices." Under review at *IEEE Transactions on Mobile Computing*.

[5] **Shizhao Gao**, Jun Li, Qiming Li. "Remember Gravity, Forget the Camera: AEGIS for View-Robust Skeleton Action Recognition." Submitted to *AAAI 2027*.

[6] **Shizhao Gao**, Jun Li, Qiming Li. "Anomaly Is Relational: Gated Social Dynamics for Multi-Person Skeleton Anomaly Detection." Submitted to *AAAI 2027*.

[7] Jun Li, **Shizhao Gao**, Tianwen Hu, Qiming Li. "FG-Net: Adaptive Node Interaction in a Flexible Graph-Based Network for Repetitive Action Counting." Under review at *IEEE Transactions on Circuits and Systems for Video Technology*.

Research Software, Mentoring and Academic Service

- **Lab infrastructure:** Developed a server management system and an **LLM API aggregation platform** for the research group; maintained shared GPU servers, coordinated compute allocation, and resolved environment issues.
- **Edge inference:** Built the Nano-TCC deployment pipeline from PyTorch through ONNX to TensorRT FP16 on Jetson Orin Nano 8 GB; sustained approximately **35 FPS** and completed a **four-hour stability test** without crashes.
- **Mentoring:** Guided undergraduate thesis projects and research interns in problem formulation, implementation, experiment planning, and academic writing.
- **Service and communication:** Reviewer for *Image and Vision Computing* (2026); presented *Count What Repeats* as a poster at CVPR 2026 in Denver, USA.

Professional Experience

Fujian Tianhe Zhongcheng Technology Development Co., Ltd. Oct. 2024–Apr. 2025

Product Management Intern

Translated bank e-guarantee workflows into requirements and acceptance criteria; coordinated development, testing, and business validation to deliver the project **one month ahead of schedule**.

Wuhan Yangtze Computing Technology Jun. 2022–Jan. 2024

Product Engineer

Supported server configuration, platform adaptation, and validation across **100+ enterprise projects**; led high-density server product introduction, contributing to **RMB 30 million** in initial orders.

Honors and Awards

- **CVPR Compute Gold Star**, 2026.
- **Academic Scholarship, Chinese Academy of Sciences**, 2025.
- **National Third Prize**, "Huawei Cup" China Graduate Mathematical Contest in Modeling, 2025 (team leader).
- Undergraduate academic scholarships and provincial awards in innovation competitions.

Technical Skills

Languages and frameworks: Python, C++, PyTorch, Transformers, NumPy, Pandas, OpenCV, MMAction2.

VLM research: Qwen3-VL, Qwen2.5-VL, LLaVA-Video, InternVL; video QA, test-time scaling, confidence-based routing, cross-model evaluation, input-token accounting, and FLOP analysis.

Systems: Linux, Git, Docker, CUDA, ONNX, TensorRT; GPU resource management and API integration.

English: Academic reading and writing, technical documentation, and everyday communication.

References

Both referees: Quanzhou Institute of Equipment Manufacturing, Haixi Institute, Chinese Academy of Sciences.

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Assoc. Prof. Qiming Li | Research supervisor
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